

Assess learning and plan next steps

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Complete: 7,305,019 rounded to 100,000; $2,407 \times 38$; $7,488 \div 26$. Copy or complete the first step.

2. Find the HCF of 72 and 120, then calculate $18 + 6 \times 7$.

3. Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain?

4. Miro says: Miro writes 'more practice' as a next step without naming the knowledge needed. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Complete: 7,305,019 rounded to 100,000; $2,407 \times 38$; $7,488 \div 26$. Copy or complete the first step.
Answer 7,300,000; 91,466; 288.
2. Find the HCF of 72 and 120, then calculate $18 + 6 \times 7$.
Answer HCF 24; result 60.
3. Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain?
Answer 2,325 items.
4. Miro says: Miro writes 'more practice' as a next step without naming the knowledge needed. Is this correct? Explain.
Answer A useful next step names the exact concept, example and support required.

Assess learning and plan next steps

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Complete: 7,305,019 rounded to 100,000; $2,407 \times 38$; $7,488 \div 26$.

6. Check this result using an inverse, estimate or known fact: Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain?

2. Find the HCF of 72 and 120, then calculate $18 + 6 \times 7$.

7. Choose one autumn objective and create a diagnostic question that would reveal a common misconception.

3. Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain?

8. Identify and correct this misconception: Miro writes 'more practice' as a next step without naming the knowledge needed.

4. Solve this independently and show every stage: Complete: 7,305,019 rounded to 100,000; $2,407 \times 38$; $7,488 \div 26$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Find the HCF of 72 and 120, then calculate $18 + 6 \times 7$.

10. Write and solve a similar question to: Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain?

Teacher answers and checking prompts

1. Complete: 7,305,019 rounded to 100,000; $2,407 \times 38$; $7,488 \div 26$.
Answer 7,300,000; 91,466; 288.
2. Find the HCF of 72 and 120, then calculate $18 \div 6 \times 7$.
Answer HCF 24; result 60.
3. Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain?
Answer 2,325 items.
4. Solve this independently and show every stage: Complete: 7,305,019 rounded to 100,000; $2,407 \times 38$; $7,488 \div 26$.
Answer 7,300,000; 91,466; 288.
5. Use a second method or representation for: Find the HCF of 72 and 120, then calculate $18 \div 6 \times 7$.
Answer Expected result: HCF 24; result 60. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain?
Answer Expected result: 2,325 items. A valid check must be shown.
7. Choose one autumn objective and create a diagnostic question that would reveal a common misconception.
Answer Answers vary. The question must target a specific objective and explain the likely error.
8. Identify and correct this misconception: Miro writes 'more practice' as a next step without naming the knowledge needed.
Answer A useful next step names the exact concept, example and support required.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Assess learning and plan next steps

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Choose one autumn objective and create a diagnostic question that would reveal a common misconception.

6. Create a new example that tests the same idea as: Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain?

2. Prove the answer to this question, rather than only calculating it: Complete: 7,305,019 rounded to 100,000; $2,407 \times 38$; $7,488 \div 26$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Find the HCF of 72 and 120, then calculate $18 + 6 \times 7$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 2,325 items.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro writes 'more practice' as a next step without naming the knowledge needed.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Choose one autumn objective and create a diagnostic question that would reveal a common misconception.
Answer Answers vary. The question must target a specific objective and explain the likely error.
2. Prove the answer to this question, rather than only calculating it: Complete: 7,305,019 rounded to 100,000; $2,407 \times 38$; $7,488 \div 26$.
Answer Expected result: 7,300,000; 91,466; 288. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Find the HCF of 72 and 120, then calculate $18 + 6 \times 7$.
Answer Expected result: HCF 24; result 60. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 2,325 items.
Answer One valid reconstruction is: Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro writes 'more practice' as a next step without naming the knowledge needed.
Answer A useful next step names the exact concept, example and support required.
6. Create a new example that tests the same idea as: Solve: 25 boxes hold 144 items each. 1,275 items are used. How many remain?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.